
22. AI in policing and law enforcement

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INTRODUCTION

One late October 2015 in Manchester, New Hampshire, Connor Deleire waited in his friend's car for him to return. What he did not know was that the car was parked in a "predictive hot spot", which is why he was singled out by the police. Not being able to give a legitimate reason for being there he became victim of a violent police arrest, as journalist Shane Kavanaugh describes (2017). Stories like this keep surfacing in public discussions about predictive policing. Official accounts of predictive policing summarize it as the use of AI or algorithmic tools to "harness the power of information, geospatial technologies and evidence-based intervention models to reduce crime and improve public safety" (National Institute of Justice 2014). There is a trend to summarize the upsurge of algorithms and machine learning in predictive policing under the notion of Artificial Intelligence (AI) (e.g. McDaniel and Pease 2022). Since AI is an ambiguous concept, this chapter follows the understanding of AI suggested in this book's introduction: a "cluster of technologies" that is socially embedded. This chapter will focus specifically on the use of *databases* and *algorithms* which, together with humans, matter in predictive policing and other algorithmic tools for law enforcement.

Originally a US-based approach, variants of predictive tools emerged, for example in China (e.g. Sprick 2020), Iraq (e.g. Perry et al. 2013), India (e.g. Marda and Narajan 2020), and Europe, including the UK (e.g. Mohler et al. 2016), Germany and Switzerland (e.g. Egbert and Leese 2021) and the Netherlands (e.g. Meijer et al. 2021). In most cases it involves the services of private companies that develop the design of such tools, structure processes (such as the flow of data required for the tool to work) and provide the software. One go-to provider for these services is the US-based company Palantir¹ that specializes in predicting patterns of most kind. In Norway, for example, Palantir was hired to develop a local version of their "super-weapon" (Gjerding and Andersen 2017) for police use. The system "omnia" literally sought to integrate any available data in Norway's landscape of public management, ranging from addresses, health data, transport, data on sentencing, pictures, to a broad range of biometric information, including DNA, in order to "upgrade" law enforcement (Trædal 2017). Palantir's project, however, was discontinued in Norway without success after 11 years and investments of 100 million Norwegian kroner (Røise 2020). The reasons for this are unclear: did such a large-scale system simply not meet Norway's social realities? Was there too little data for the tool to work efficiently? Or did the emerging European pushback initiated by anti-discrimination initiatives and civil liberty unions factor into the decision (e.g. Fair Trials 2022)?

This chapter will dive into these issues and others when it explains the assumptions, decision-making processes and technological models that drive predictive policing today. It will summarize how predictive processes and techniques are discussed and describe how they materialize and are made to matter in policing (Kaufmann 2023). What is more, the chapter will also provide a brief outlook into new areas of prediction within policing. It brings the discussion into the field of genomic predictions for crime investigation. Here, algorithms are

used to derive predictions about genealogy (familial relations) and physiognomy from DNA in order to identify suspects. The chapter will focus on the latter. This field is chosen, because it no longer relates to the “territory” of geographic models, but it ties prediction to suspicious groups based on biological factors. To this extent, it is a technological practice that exacerbates the already sensitive context of law enforcement in which predictive technologies are used.

STATE OF THE ART: DEBATES ON PREDICTIVE POLICING

Considering the history of predictive techniques, Connor Deleire could have had the same experience without the use of algorithms. The origins of predictive policing are analogue, where police data were translated into hotspots marked with pins on a map. Such techniques were part of the project to scientize police patrol (Manning 2008) and gave rise to the idea of evidence-based policing (Sherman 1998). Rational Choice Theory (Cornish and Clarke 1986) is one of the first and best-known theoretical attempts to conceptualize projects of this kind. The theory is based on the assumption that offenders follow patterns of spatial and temporal proximity of their original offence when they commit the next crime, because they are familiar with the scene and local habits. The Near-Repeat Hypothesis (e.g. Townsley et al. 2003) emerged in relation to this model and was tested in the field of domestic burglary. Proponents of predictive policing, then, argue that such patterns are useful to focus police efforts. Officers would be able to focus attention on crime-prone areas (Kennedy et al. 2011). According to these expectations, predictions would render police work more efficient and provide relief for overworked police officers – all based on statistical knowledge (Beck and McCue 2009). This logic also corresponded with managerialist reforms of police work and the rising focus on cost-efficiency. Digitization, then, enabled the relaunch of “geographically applied criminology” (cf. Kaufmann et al. 2019), because one could include more data, more parameters (population density, events, architecture and environmental characteristics, weather forecasts, etc.), as well as sophisticated algorithmic models to find patterns. Not least, digitization provided a certain degree of automation. This development is welcome among some communities, because it bears the promise of enhancing the precision of prediction and providing speedy, actionable results. Efficient information processing and a prioritization of police actions are the main drivers for the use of algorithms for predictive policing (*ibid.*).

The development of these models, related technocratic approaches and the arguments for adopting these tools do, however, meet criticism. Critique is advanced, for example, by police professionals who are sceptical of algorithmic solutions (Sandhu and Fussey 2021). Their doubt includes ethical and moral concerns (Gundhus et al. 2017). Officers also point to challenges such as technology failures (Kaufmann 2018) and cost-inefficiencies. Further, some officers feel insecure due to a lack of digital literacy or fear that their competences may be replaced by automated systems (*ibid.*). A more fundamental concern relates to the role of the state and democratic accountability (for a systematic review of this concept in the context of AIT uses, see Cobbe and Singh, Chapter 7 in this volume). Mark Andrejevic argues, for example, that predictive policing nurtures a logic of “data collection without limits” (2018), which leads to a general increase in surveillance. Linked to that problem is the critique that only those with access to data and processing power will be able to create and access insights (Andrejevic and Gates 2014). Especially in the sensitive context of crime governance, such practices enhance already existing power imbalances. Limited access to and knowledge of databases and math-

ematical models is a central problem. This, paired with the fact that most software solutions involve commercial players, leads to a lack of transparency that is difficult to address (Meijer and Wessels 2019). Many solutions are owned by private companies and a close collaborative process between companies and clients is not necessarily a given. Sometimes algorithmic designs are even treated as business secrets. This lack of transparency and the multitude of stakeholders involved in the process, again, entail problems of accountability (Bennett Moses and Chan 2018; for a discussion of transparency principles in AIT uses, see Aula and Erkkilä, Chapter 13 in this volume). Accountability is, however, particularly relevant as the status of neutrality that algorithmic technologies enjoy is challenged by a variety of problems at a systemic level (for a critique see Wilson 2018). Feedback loops (created by sending police officers back to the already well-policed and stigmatized areas) shape databases, which leads to a perpetuation of social imbalances and injustices (Gill 2000; Harcourt 2007; Brayne 2017). The result is over-patrolling of poor and minority neighbourhoods (Marda and Narayan 2020). This problem interlinks with racial biases and skewed datasets that create the basis for algorithmic predictions (Ferguson 2017; Richardson et al. 2019), which leads to *systematic* forms of unfair and discriminatory treatment (Alikhademi et al. 2022). Beyond that, the social life of a prediction, which is how and when officers take action based on them, could also produce problems (Richardson et al. 2019) as we saw in the case of Connor Deleire. Many problems of bias are dependent on the software type used, the databases involved to train them and human-decision making.

UNDERSTANDING HOW PREDICTION ALGORITHMS COME ABOUT AND CO-CREATE SOCIETY

Using predictive technologies in law enforcement is part of a larger trend towards a “pre-crime society” (Zedner 2007; Mantello 2016). This trend implies an epistemological shift: the problem of criminality is relocated into the future with the ambition of governing crime before it occurs (this is also true, for example, when predicting child abuse with risk factors: Krutzinna, Chapter 29 in this volume). Prediction and prevention are epitomes of this shift, where the problem of “prepression” (repression as a result of prediction, Schinkel 2011) and a drift towards crime preemption emerge, too (Egbert and Krasmann 2020). Within this pre-crime approach in public policy, large databases and algorithmic calculation engender a governance by patterns and categorizations that are the basis for predictions. Patterns and categories, however, perform their own politics (Kaufmann et al. 2019). In order to serve as an argument for intervention, patterns reduce, simplify and discriminate (in the sense of *distinguishing entities from each other* and *discriminating against*). The decision-making processes that inform these patterns are a part of complex technological developments and remain largely invisible. This underlines the relevance of studying the social life of prediction algorithms, their emerging agencies (see the introductory chapter to this volume), as well as the collaborations between humans, datasets and machines in creating predictions (Kaufmann 2018; 2023), to which we turn now. Already in the early 1990s Stephen Ackroyd et al. pointed out that the integration of policing with information systems is shaped by many aspects, such as “the nature of that work, how it is organized day-to-day, what tacit understandings are built into this organization, its situatedness within a network of other organizational arrangements” (1992: 26).

Following the *algorithm's life cycle* is one approach to study the functionalities, different forms of agency and politics of algorithms in context (ibid.). Tracing a life cycle allows researchers to determine those points at which important design decisions are taken (which often cascade through the entire prediction process), but also those moments where algorithms exhibit agencies that influence predictions (Kaufmann and Leese 2021). Indeed, the concept of the life cycle carries the ontology of data and algorithms as lively (Lupton 2016) and dynamic (Barad 2003). As opposed to analytics that focus on the role of humans in “mastering” technologies, the life cycle draws a more complex picture of predictive processes. It suggests that things, including algorithms, live their own life and intersect with the many life cycles of others in processes of mattering (Kaufmann 2023). Together, humans, data and technologies also define whatever may be considered efficiency and situations that require action by law enforcement representatives. The remainder of this chapter is used to sketch out a prototype life cycle of a predictive policing algorithm, before giving an outlook into a new area for predictions in law enforcement that involve DNA.

THE LIFE CYCLE OF AN ALGORITHM IN PREDICTIVE POLICING

An algorithm's life begins before it is written, namely with *imaginaries* and preconceptions about its purpose and abilities. The purpose of the algorithm is linked to the data that developers believe are best to train algorithms. Ruha Benjamin (2019) addresses such imaginaries when she writes about names as data. What lies in a name comes with an expectation or association, in the worst case a prejudice. If software developers believe that names can capture a group's heritage or a type of neighbourhood, names would be used as data with the aim of distinguishing neighbourhoods for prediction.

Imaginaries also drive *data generation, curation and cleaning*. Already the ways in which data is first collected, according to specific protocols and on-site routines, are integral to the dataset which is later used to train algorithms. In predictive policing, for example, one would need data about crime. Already the (analogue) form that police officers fill out determines what type of information will be available for the algorithm's training. This analogue form is embedded in social and policing processes. Not only do forms express police protocol that has developed over many years, but also incorporate ideas about what data are needed for police work. In most cases, such data are generated with the intention of a *specific* usage (via registration or surveys). Other datasets were generated with a different or more general aims in mind (e.g. data about number plates), but are reused for the purpose of creating predictions (cf. Palantir's approach).

This illustrates that *data storage* influences predictive tools, too. In predictive policing we find a broad spectrum that starts out with small, highly cleaned datasets that are chosen according to specific theories (Kaufmann 2019). When tools, for example, draw on Townsley's Near-Repeat Hypothesis, predictions would be based on data that capture where, when and what type of crime occurred. Such data may be extracted from existing datasets or are collected during police patrol. At the other end of the spectrum we find approaches that are interested in most types of stored data from weather data to population density. In that case, data from many public and private databases are combined, because the approach is to create algorithms that correlate “very big” datasets with each other in order to identify new types of patterns (this approach is also reflected in the notion of “interoperability”, see Leese, Chapter

11 in this volume). Here, the role of commercial databases makes a difference since their rules for storage tend to be less strictly regulated. However, as the abovementioned example of Palantir's "omnina" showed, big data predictions can also include public sector databases. A known and obvious challenge are datasets that are based on discriminatory data collection practices (cf. Magnet 2011). Such biases are then carried throughout the entire design of the algorithm (Mann and Matzner 2019). The fact that data cascade and influence predictions illustrates how datasets assume agency. Other examples of the agencies of datasets are the moments at which algorithms identify surprising patterns, for example when they start to predict crime in accordance with phases of the moon (Kaufmann 2019). This pattern may be spurious, or it may predict well. *Why* the pattern would predict well is not necessarily explored. In such models, theory and causality are no longer important as long as actionable patterns are produced. This goes to show that the dataset used to train algorithms has epistemological implications: some datasets are used to predict according to a specific theoretical model. A different approach is to "let the data speak for themselves", that is to infer new patterns without being guided by a specific hypothesis.

For the algorithm to *emerge* police officers and developers must select content parameters (such as *time*, *space* and *type of crime* for algorithms based on Near-Repeat models). The team would first have to agree on the type of data that expresses these parameters. Such parameters are then translated into mathematic form. Though the expertise of those writing the algorithm's mathematic "code" is central at this stage, it is not a given that those who define content parameters and those who translate them into mathematic variables understand each other (Kaufmann 2018). Here, decision-making processes and frictions can occur that determine the entire process all the way to the algorithm's outputs.

Predictive policing algorithms emerge when they are trained. One such (simplified) training process could look like this: the mathematic parameters are put to use on datasets where the correct hits for a pattern are known. Developers follow the correlation process and give feedback about which result is considered an actual result. Note that giving feedback is here also an influential decision-making process, where "correct" hits based on (potentially biased) databases and value statements guide the training of the algorithm. When the algorithm identifies a pattern that is considered to be wrong, the parameters are adjusted. Here, algorithms can provoke frictions, for example when they identify ambiguous patterns (Kaufmann 2019).

The algorithm is considered *mature* when it has become good enough at identifying the results that match the algorithm's predefined aims and the patterns considered a correct hit. It can now be run on datasets where the "correct" hits are not known. Its agency becomes very pertinent at this stage as it now provides new insights to police officers the origins of which are no longer easily traceable. Algorithms do not disclose the arguments about how they have reached a specific result, which complicates the accountabilities involved when taking action based on algorithmic results. What is more, algorithms can now start to delete parameters that they identify as irrelevant. These concrete influences do change police cultures as they literally guide police patrol and foster new knowledge practices based on patterns (Kaufmann 2019).

What happens to prediction algorithms at the end of a life cycle? Do they reach a natural limit, are they ever retired? Whether algorithms *die*, that is, whether they will stop computing or identifying predictions, is mathematically speaking an unknown. The answer to this question is not computable (ibid.). The respective algorithm has to be run. However, whether the algorithm stops computing or not, whether it is taken out of work or not, is not necessarily tied to its capacity to act. Their agency is still present in decision-making. Training data or param-

eters can also be reused in new policing algorithms, venturing from one context to the next. Such instances of reuse raise new concerns about systemic biases that cascade across systems. The death of an algorithm would be the moment at which all of its elements no longer matter, which is rarely the case.

Following the life cycle of a predictive policing algorithm can assist in the identification of key moments at which collaborations occur that impact the entire prediction process, including the ethics, politics and values predictions perform.

Algorithmic policing technologies are not reduced to geographic predictions. Algorithms are also used in crime investigations to create predictions about familial relations and physical appearance from DNA. In the next part I will focus on the latter.

THE LIFE CYCLE OF AN ALGORITHM IN DNA ANALYSIS

In the forensic landscape of the late 1980s DNA emerged as a competitor to fingerprinting. Contrary to its role in the criminal justice system today, the suitability of DNA for forensic investigation was so contested that proponents borrowed authority from fingerprinting to argue for the evidentiary value of DNA (see *DNA fingerprinting*; Aronson 2007). The main aim of using DNA in forensics, then, was to establish matches between two DNA samples – one from a crime site, one from a database – for the sake of identification.

In order to match two samples, DNA first needs to be sequenced. Sequencing is to prepare DNA samples in a way that allows forensic scientists to identify four regions of the genome (so-called Short Tandem Repeats, STRs) that distinguish individuals from each other. Over the years, this number of regions rose to over 24 STRs, which means that by today, DNA profiles are more comprehensive, changing the possibilities for finding a match. A second important factor for identification is the growing availability of databases. Without the storage of DNA data, there would be no option to compare samples. A third factor that influences the identification of matches concerns the use of statistics to express the likelihood of two DNA samples stemming from the same person. In the struggle to establish procedure for DNA use, debates among the forensic and legal community revolved around the statistical forms and models used to express the likelihoods (Gigerenzer and Hoffrage 1995; NRC 1992). Different statistical models would lead to considerably different results, which had a direct impact on sentencing (cf. Lynch et al. 2008). This example emphasizes that any use of new data analyses in law enforcement sparks discussions about the “correct” analytic models, each of which creates different societal impacts – as we have also seen in predictive policing.

This brief history of DNA analysis in law enforcement is important to appreciate that DNA samples first had to become a trusted and reliable source for knowledge production in the law enforcement system. For these imaginaries and expectations to emerge, concrete scientific discourses and framing efforts were undertaken (Lynch et al. 2008). Only when these were established would DNA collection and analysis become a viable policy. Nowadays, DNA is considered the “gold standard” of forensic science. DNA has become a “truth machine” (ibid.). This *imaginary* heavily influences the use of DNA in investigations and in court today. What is more, it is also a driver for scientific endeavour to develop DNA analysis further and integrate it with more advanced computational practices. Algorithms are used at many stages and for many different reasons in DNA analysis today (to create profiles, analyse DNA mixtures, calculate probabilities, etc.). In what follows I focus on the use of algorithms for phenotypic

predictions that increasingly inform police work. Phenotyping is the process of predicting physiognomic characteristics from a DNA sample.

An important moment in the rise of such predictions was the arrival of Next-Generation Sequencing (NGS), also called Massive Parallel Sequencing (MPS). These sequencing techniques no longer focus on STRs, but on other regions of the genome. These are also called “coding regions”, because they code for proteins, which again shape gene expression. Coding regions are, for example, analyzed to identify proneness to illnesses, but they are also increasingly integrated with forensic work. As the names for the technique signal: these sequences contain different, and *much more*, information than STR profiles. The expressions “Massive” and “Next Generation” capture this considerable increase in information. Hence, this shift has been described as the beginning of “big genome data” (Murphy 2018) – the rise of genomic data so large that it has to be analyzed with more complex algorithms.

As we have seen above, the functionality of algorithms is highly integrated with the data and data imaginaries they analyze. Together, they form a “cluster of technologies”. In order to understand the social life of an algorithm for DNA analysis, one needs to already take account of DNA sampling, because it creates the base data for analysis. DNA sampling, for example, involves a lot of “low tech” equipment, such as analogue forms that give context about the origin of the trace containing DNA. In the laboratory, traces are extracted and apparatuses such as centrifuges and thermo-regulators prepare DNA for the subsequent range of biochemical processes which eventually result in a sequence. This sequence eventually appears as a code on a computer screen. As illustrated with the case of predictive policing, these early steps are relevant, because they create the data to be stored and analyzed. The development of protocol for DNA sampling is also always a negotiation between humans and technologies.

Now that a DNA sample is sequenced and exists as digital data, it can be integrated with computation. In order to train and use algorithms to analyze DNA and create predictions, sequences have to be *stored* at a large scale. Some DNA databases are owned by the National Crime Investigation Services or different law enforcement institutions, others are owned by forensic departments in hospitals, or commercial actors. In relation to STR datasets Erin Murphy found that the storage and categorization of DNA data varies across institutions, and data-exchange between different database providers is not necessarily streamlined (Murphy 2018). Databases, then, emerge as “inventive, sometimes unruly” sites (Nadim 2016: 516), where data work, such as negotiations about the type of information stored and cleaning processes are needed to create consistent datasets. Yet, even if the algorithm is trained on one specific dataset, it is not a given that these data stem from a database where storage and curation is streamlined. Such socio-technical processes also need to be taken into account when building databases with NGS profiles.

Datasets containing DNA profiles and details about their owners can now be used for *training* algorithms that predict physiognomic features from DNA. This process is called phenotyping. Phenotyping software is advocated as an investigative tool of use to the police when a “classic” STR profile does not yield a match. The company Parabon, for example, offers a software called “Parabon Snapshot” that predicts a “typical” facial image from DNA, similar to a classic identikit picture. Officers would then be able to use such predictions to focus their investigations on a group with specific physiognomic expressions. Parabon is, however, a commercial solution and not yet broadly accepted among police officers.

The training of an algorithm to predict a facial image based on DNA is not too dissimilar from that for predictive policing described above. I will draw on Rose Hopman’s work to

explain the process. In order to estimate eye, hair and skin colour, genes known to impact pigmentation are correlated with DNA profiles of individuals whose geno- and phenotype is known (Hopman 2021). The algorithm, then, correlates pigmentation genes with physiognomic expressions. From these correlations, it is inferred which genomic pattern expresses what type of pigmentation. The size and the quality of the database decides, however, what type of correlations can be established and what patterns can be found. The next step is then to link the continuous reading of colour (a scale that is not subdivided into colour categories), to grades of, for example, skin colour for the sake of prediction (Hopman 2021). This work of categorizing skin colour as “white”, “intermediate” and “black” is done by humans, who perform a highly political act, because these grades express generalizations that are now part of identification processes (*ibid.*).

In order to complete the “snapshot” image, a similar process is used to predict the morphology of the face. Here, a dataset is needed featuring records of individuals whose facial morphologies and genomic profiles are known. Biometric imaging is used to capture facial morphologies. These images are then standardized and symmetrized, which involves the cleaning of irregularities in order to render the morphologies comparable (*ibid.*). This is again a highly political process, because it involves templates of human “types”. These morphological types are then linked to similarities in the respective genomes. Eventually, the algorithm *emerges* when it integrates these two pattern-based approaches to predict the facial features of a DNA owner. Again, not only the available datasets, but also the processes of categorizing skin colour and “cleaning away” morphological “deviances” are practices that influence the prediction. Most phenotyping instruments still focus on eye, hair and skin colour, with new features being added throughout. It is mainly the product Parabon Snapshot (Parabon, n.d.) that already offers a prediction of the entire face.

Phenotyping of eye, hair and skin colour is used as a tool for intelligence in the US and in an increasing number of European countries, which represents an epistemological shift. DNA is no longer used to express a likely match, but analyzed to offer physiognomic expressions to guide investigations. However, the use of phenotyping is debated. Arguments about new possibilities for efficient intelligence are countered by concerns about reliability and discrimination. The above life cycle illustrates that outputs are highly dependent on a range of different actors that influence prediction processes; predictions involve biased databases that emerge from discriminatory policing practices (e.g. Skinner 2020) and concrete value statements about physical appearance (Hopman 2021) (for wider discussion of bias: see Hong, Chapter 8 in this volume). The intention of phenotyping is no longer to ascertain an individual’s identity, but to enlarge the pool of suspects from which new investigative leads can be inferred. Phenotypic predictions, then, offer physiognomic predictions in order to draw groups with a specific physiognomic profile into the realm of suspicion (Hopman and M’charek 2020).

While some of these criticisms remind us of the debates about geospatial predictions, phenotyping introduces an entirely new level of challenges, because it generates predictions about groups of people based on their physical appearance. Moreover, the involvement of DNA represents a new form of data capture and biometric surveillance that opens the door for entirely new policing practices.

CONCLUSION

The histories of public policy, policing and their relationship to information technologies are more integrated than one may expect. From the fifteenth to the seventeenth century “polizey” represented the moral oversight in a state and was derived directly from the concept of “policy” (Weatherby 2016). Before public policy and the police became established as more distinct fields, they already organized many areas of life by aid of information (*ibid.*). Statistics, for example, were relevant instruments for administration and governance back then and today (*cf.* Foucault 1977–78).

While there are continuities in doing (public) administration by means of counting, calculating and data processing, the rise of large datasets and new tools for data processing introduced qualitative shifts – also in fields that we define as policing today. This chapter has documented how the “clusters of technology” involved in predictive policing and DNA phenotyping create new epistemological practices in law enforcement. Predictive instruments strengthen the role of patterns in the governance of crime. While geospatial patterns define vulnerable areas to be policed, physiognomic patterns are used to direct crime investigations. The generation of patterns is a highly complex process that combines a vast range of decisions and agencies. What is more, in their function as actionable, “efficient” knowledge patterns also exercise their own agency. They guide the police to vulnerable areas and people of a specific physical appearance. The creation of territorial markers or pheno-*types* of potential suspects plays into the desire for efficient, technologically upgraded and evidence-based policing. Yet, these patterns hide the imaginaries, practices and hypotheses that guide the adoption of new technologies, data collection, the decisions that direct data generation and cleaning, the structural designs and inconsistencies of databases, the parameters that are defined to train algorithms and the mathematic models that guide analyses.

All of these elements determine the knowledge algorithmic technologies create and the politics they exercise (Kaufmann et al. 2019). Identifying and addressing these relationships is particularly relevant in the sensitive domain of law enforcement, where predictions produce concrete security and juridical implications.

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NOTE

1. Other known companies are Predpol (n.d) which bought Australian HunchLab, Azavea (n.d.) and Precobs (Institut für musterbasierte Prognosetechnik n.d.).

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